

LEVEL-I (MAIN)

EXERCISE-I

Straight Objective Type Questions

(d-block and general characteristics)

- Which set of elements among the following are called non-transitional elements
1) Cu, Ag and Au 2) Fe, Co and Ni 3) Zn, Cd and Hg 4) Re, Os and Ir
- The following represents the electronic configuration of a transition element
1) $ns^2 np^3$ 2) $ns^2 np^6 nd^3 (n+1)s^2$
3) $ns^2 np^6 nd^{10} (n+1)s^2 (n+1)p^4$ 4) $ns^2 np^5$
- Outer electronic configuration of the element Palladium is
1) $4d^5 5s^1$ 2) $4d^9 5s^2$ 3) $4d^{10} 5s^1$ 4) $4d^{10} 5s^0$
- The following pair of ions have the same electronic configuration
1) Cr^{3+} and Fe^{3+} 2) Fe^{3+} and Mn^{2+} 3) Fe^{3+} and Co^{3+} 4) Sc^{3+} and Cr^{3+}
- The atomic numbers of vanadium chromium, manganese and iron are respectively 23, 24, 25 and 26. Which one of these may be expected to show highest second ionization enthalpy
1) V 2) Cr 3) Mn 4) Fe
- Which of the following element has high density
1) Sc 2) Zn 3) Os 4) Co
- Of the following outer electronic configurations of atoms, the highest oxidation state is achieved by which one of them?
1) $(n-1)d^8 ns^2$ 2) $(n-1)d^5 ns^2$ 3) $(n-1)d^3 ns^1$ 4) $(n-1)d^5 ns^1$
- The common oxidation state of transition elements is
1) + II 2) + IV 3) + VI 4) + VII
- Coloured complexes absorb radiation in the
1) visible region 2) infrared region 3) U.V. region 4) far IR region
- Coloured ion among the following is
1) Zn^{2+} 2) Mn^{2+} 3) Cu^{1+} 4) Ti^{4+}
- Which of the following pairs of ions are colour-less ?
1) Ti^{+3} , Cu^{+2} 2) Sc^{+3} , Zn^{+2} 3) Co^{+2} , Fe^{+3} 4) Ni^{+2} , V^{+3}
- The following ion exhibits highest magnetic moment ?
1) Cu^{2+} 2) Ti^{3+} 3) Ni^{2+} 4) Mn^{2+}
- (A) : The spin only magnetic moment of Sc^{+3} is 1.73BM
(R) : The spin only magnetic moment (in BM) of an ion is equal to $\sqrt{n(n+2)}$
1) Both (A) and (R) are true and (R) is the correct explanation of (A)
2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
3) (A) is true but (R) is false 4) (A) is false but (R) is true
- The metal not present in German silver is
1) Ag 2) Cu 3) Ni 4) Zn

15. (A) : Transition elements form alloys easily
(R) : Transition elements have high melting points
16. M^{3+} ion of the first transition series metal 'M' has a magnetic moment 1.73 BM. The atomic number of the metal 'M' is
1) 21 2) 24 3) 29 4) 22
17. Which of the following is an interstitial compound
1) NH_3 2) $PdCl_2$ 3) PdH 4) SiC
18. The type of bond expected to form atoms of two elements in interstitial compounds is
1) Covalent 2) Ionic 3) Metallic 4) None
19. CrO_4^{2-} and MnO_4^- ions exhibit colour due to
1) presence of unpaired electrons in 'd' orbitals of Cr and Mn
2) charge transfer phenomenon
3) d-d electron transition 4) close packing crystal structure
20. Ti^{3+} is purple but Ti^{4+} is colourless. This is because
1) d^1 configuration of Ti^{3+} 2) d^0 configuration of Ti^{3+}
3) d^3 configuration of Ti^{3+} 4) d^{10} configuration of Ti^{3+}
21. Which of the following set of elements are Ferromagnetic in nature
1) Zn, Cd and Hg 2) Cu, Ag and Au 3) Fe, Co and Ni 4) Sc, Ti and U
22. The following is paramagnetic
1) $CaCl_2$ 2) $CuCl_2$ 3) $ZnCl_2$ 4) $NaCl$
23. Ferrous ion changes to 'X' ion, on reacting with acidified hydrogen peroxide. The number of d-electrons present in 'X' and its magnetic moment are
1) 6 and 6.93 B.M. 2) 5 and 5.92 B.M. 3) 5 and 4.9 B.M. 4) 4 and 5.92 B.M.
24. Alloy formation ability of transition elements is due to
1) same crystalline structures 2) same atomic radii
2) similar chemical properties 4) Both 1 & 2
25. The pair of compounds in which metals are in the highest possible oxidation state is
1) $[Fe(CN)_6]^{3-}$, $[Co(CN)_6]^{3-}$ 2) CrO_2Cl_2 , MnO_4^- 3) TiO_2 , MnO_2 4) $[Co(CN)_6]^{3-}$, MnO_2
26. The melting point of transition elements is very high due to
1) Involment of greater no. of electrons in (n-1)d and ns electrons in the inter atomic metallic bonding
2) Non-involment of greater no. of electrons in (n-1)d and ns electrons in the inter-atomic metallic bonding
3) Involment of greater no. of electrons in 'ns' electrons in the inter-atomic metallic bonding
4) Non-involvement of greater no. of electrons in 'ns' electrons in the inter-atomic metallic bonding
27. The catalyst used in the reduction of perdisulphate to sulphate with iodide ion is
1) Fe^{3+} 2) Fe^{2+} 3) Both 1 and 2 4) Cr^{2+}
28. Amongst the following species, maximum covalent character is exhibited by
1) $FeCl_2$ 2) $ZnCl_2$ 3) $HgCl_2$ 4) $CdCl_2$

preparation and properties of compounds :

29. Which the following statements is wrong ?

- 1) An acidified solution of $K_2Cr_2O_7$ liberates iodine from iodides
- 2) In acidic solution dichromate ions are converted to chromate ions
- 3) Ammonium dichromate on heating undergoes exothermic decomposition to give Cr_2O_3
- 4) Potassium dichromate is used as titrant for Fe^{+2} ions

30. In the dichromate dianion

- 1) 4 Cr – O bonds are equivalent
- 2) 6 Cr – O bonds are equivalent
- 3) All Cr – O bonds are equivalent
- 4) All Cr – O bonds are non-equivalent

31. When conc. H_2SO_4 is added to solution of potassium dichromate

- 1) A yellow solid of chromate sepearates out
- 2) A yellow solid of Cr (VI) oxide sepearates out
- 3) A purple red solid of chromate separates out
- 4) A purple red solid of Cr (VI) oxides sepearates out

32. Which one of following does not decolourise an acidified $KMnO_4$ solution ?

- 1) SO_2
- 2) $FeCl_3$
- 3) H_2O_2
- 4) $FeSO_4$

33. When conc. H_2SO_4 reacts with $KMnO_4$ an explosive compounds is obtained which is

- 1) MnO_2
- 2) Mn_2O_7
- 3) MnO_4^{-2}
- 4) None

34. When Na_3MnO_4 is kept in strongly acidic solution the products obtained are

- 1) Na MnO_4 , MnO_2 , H_2O
- 2) $Na_2 MnO_4$, MnO_2 , H_2O
- 3) Na MnO_4 , Mn, H_2O
- 4) $Na_2 MnO_4$, Mn, H_2O

35. Addition of $K_4 [Fe(CN)_6]$ solution to $FeCl_3$ solution give

- 1) Ferro-Ferri cyanide
- 2) Ferric-Ferrocyanide
- 3) Ferric-Ferri cyanide
- 4) None

36. The brown ring complex compound is formulated as $[Fe(H_2O)_5NO]SO_4$. The oxidation states of iron is

- 1) +1
- 2) +2
- 3) +3
- 4) +4

37. Ferrous sulphate on heating gives

- 1) SO_3
- 2) SO_2
- 3) Fe_2O_3
- 4) All

38. Number of moles of $K_2Cr_2O_7$ reduced by one mole of Sn^{+2} ion is

- 1) $\frac{1}{3}$
- 2) 3
- 3) $\frac{1}{6}$
- 4) 6

39. $(X) + K_2CO_3 + \text{Air} \xrightarrow{\text{heat}} (Y)$. $(Y) + Cl_2 \longrightarrow (Z)$ Pink Which of the following is correct ?

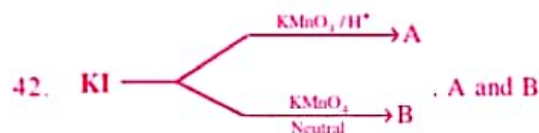
- 1) X = black, MnO_2 , Y = Blue, K_2CrO_4 , Z = $KMnO_4$
- 2) X = green, Cr_2O_3 , Y = Yellow, K_2CrO_4 , Z = K_2CrO_7
- 3) X = black, MnO_2 , Y = green, K_2MnO_4 , Z = $KMnO_4$
- 4) X = black; Bi_2O_3 , Y = colourless $KBiO_2$, Z = $KBiO_3$

40. CrO_4^{2-} (yellow) changes to $Cr_2O_7^{2-}$ (orange) in $P^H = x$ and vice versa in $P^H = y$. Hence x and y are

- 1) 6, 8
- 2) 6, 5
- 3) 8, 6
- 4) 7, 7

41. Which of the following statement(s) is (are) true for Mohr's salt ?

- 1) It does not decolourises $KMnO_4$ solution
- 2) It is a double salt
- 3) Oxidation state of Fe is +3
- 4) It is a secondary standard solution



respectively in the given reactions

- 1) I_2 , IO_3^-
- 2) IO_3^- , IO_4^-
- 3) IO_3^- , I_2
- 4) I_2O_5 , IO_3^-

43. An explosion takes place when conc. H_2SO_4 is added to $KMnO_4$. Which of the following is formed

- 1) Mn_2O_7
- 2) MnO_2
- 3) $MnSO_4$
- 4) Mn_2O_3

44. Cr_2O_3 on reaction with HCl and the product on reaction with $NaOH(aq)$ give respectively

- 1) CrO_2Cl_2 , CrO_4^{2-}
- 2) $Cr(OH)_3$, CrO_4^{2-}
- 3) Cl_2 , $Cr_2O_7^{2-}$
- 4) Cl_2 , $Cr(OH)_3$

45. Which of the following oxides of chromium is amphoteric in nature ?

- 1) CrO
- 2) Cr_2O_3
- 3) CrO_3
- 4) CrO_5

46. Which compound is deliquescent ?

- 1) Hg_2Cl_2
- 2) $HgCl_2$
- 3) $ZnCl_2$
- 4) $CdCl_2$

47. Which is coloured species ?

- 1) MnO_4^-
- 2) TeO_4^-
- 3) ReO_4^-
- 4) All

48. In the reaction $2CuCl_2 + 2H_2O + SO_2 \longrightarrow A + H_2SO_4 + 2HCl$, 'A' is

- 1) Cu_2Cl_2
- 2) Cu
- 3) $CuSO_4$
- 4) CuS

f-block elements :

49. Lanthanides like Eu and Yb can form hydrides like EuH_2 and YbH_2 . These hydrides are

- 1) Ionic hydrides
- 2) Covalent hydrides
- 3) Metallic hydrides
- 4) Intermediate hydrides

50. Lanthanoid contraction implies

- 1) Decrease in density
- 2) Decrease in mass
- 3) Decrease in ionic radii
- 4) Decrease in radioactivity

51. If the lanthanoid element with x f electrons has a pink colour, then the lanthanoid with $(14-x)$ f electrons will have the colour as
 1) Blue 2) Red 3) Green 4) Pink
52. The electronic configuration of actinoides cannot be assigned with degree of certainty because of
 1) Overlapping of inner orbitals
 2) Free moment of electrons over all the orbitals movement
 3) Small energy difference between 5f and 6d levels 4) None of the above
53. The sum of the first three ionization energies of the lanthanoides Ce, Eu, Gd, Yb, Lu is
 1) $Ce > Eu > Gd > Yb > Lu$ 2) $Yb > Lu > Eu > Gd > Ce$
 3) $Yb > Eu > Lu > Gd > Ce$ 4) $Yb > Eu > Gd > Lu > Ce$
54. The reason for the stability of Gd^{3+} ion is
 1) 4f sub shell half-filled 2) 4f sub shell completely filled
 3) possesses the electronic configuration of noble gases
 4) 4f sub shell empty
55. The +3 ion of which one of the following has filled 4f sub shell ?
 1) La 2) Nd 3) Gd 4) Ac
56. Lanthanide for which +2 and +3 oxidation states are common is
 1) La 2) Nd 3) Ce 4) Eu
57. Misch metal is
 1) An alloy of Al 2) A mixture of chromium and $PbCrO_4$
 3) An alloy of lanthanoid metals 4) An alloy of copper
58. More number of oxidation states are exhibited by the actinoids than by the lanthanides. The main reason for this is
 1) Greater metallic character of the lanthanoids than that of the corresponding actinoids
 2) More active nature of the actinoids
 3) More energy difference between 5f and 6d orbitals than between 4f and 5d orbital's
 4) Lower energy difference between 5f and 6d orbitals than between 4f and 5d orbital's

Numerical Value Type Questions

59. For how many of the following the weight increases in the applied magnetic field. NO , NO_2 , O_2 , $K_4[Fe(CN)_6]$, KO_2 , $MnSO_4$, $CuSO_4$, $ZnSO_4$
60. Permanganate ion oxidises manganous ion to manganese dioxide in presence of zinc oxide as catalyst as $MnO_4^- + Mn^{2+} + H_2O \rightarrow MnO_2 + H^+$. The number of H^+ ion added to the product side in the balanced ionic equation is
61. How many maximum bonds are equivalent in the dichromates dianion ?
62. How many incomplete orbitals are present in inner transition elements ?
63. No. of 'f' electrons in ground state of Thorium
64. Max no. of unpaired e^- in ground state of Gadolinium (Gd, $Z = 64$)

EXERCISE-II

Straight Objective Type Questions*(d-block and general characteristics)*

- The colour of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is due to
 - transfer of electron from Titanium to another atom of Titanium
 - presence of water molecules
 - d-d transition
 - intra molecular vibration
- Hydrated Cu^{2+} ions absorb light of colour and transmit light of colour
 - red and blue
 - green and purple
 - purple and red
 - blue and red
- Which of the following pair of transition metal ions, have the same calculated values of spin only magnetic moment ?
 - Ti^{+2} & V^{+2}
 - Fe^{+2} & Co^{+2}
 - Cr^{+2} & Fe^{+2}
 - Co^{+2} & Ti^{+2}
- Which of the following ions has the maximum magnetic moment
 - Mn^{+2}
 - Fe^{+2}
 - Ti^{+2}
 - Cr^{+3}
- The magnetic moment of an ion in its +3 oxidation state is 3.85 BM. The number of unpaired d-electrons present in it are
 - 2
 - 3
 - 4
 - 5
- What is the correct order of spin only magnetic moment (in BM) of Mn^{+2} , Cr^{+2} and V^{+2} ?
 - $\text{Mn}^{+2} > \text{V}^{+2} > \text{Cr}^{+2}$
 - $\text{V}^{+2} > \text{Cr}^{+2} > \text{Mn}^{+2}$
 - $\text{Mn}^{+2} > \text{Cr}^{+2} > \text{V}^{+2}$
 - $\text{Cr}^{+2} > \text{V}^{+2} > \text{Mn}^{+2}$
- Identify the correct statements among the following
 - Both Cr and Cu show +1 oxidation state
 - The complementary colour of absorbed green colour of visible radiation is purple.
 - Ni^{+2} ion in its hydrated state exhibits green colour
 - Devarda's alloy contains least percentage of 'Zn'
 - All
 - I, II, III only
 - I, IV only
 - I, III only
- Which of the following statements is incorrect in the context of paramagnetic materials ?
 - A paramagnetic material placed in a magnetic field attracts magnetic lines of force towards it
 - A paramagnetic material moves from a weaker to a stronger part of the magnetic field
 - Paramagnetic arises as a result of unpaired electron spins in the atom
 - It is difficult for magnetic lines of force to travel through a paramagnetic material than through a vacuum
- A transition metal exists in its highest oxidation state. It is expected to behave as
 - A chelating agent
 - A central metal in a coordination compound
 - An oxidising agent
 - A reducing agent

10. Ionisation energies of Ni and Pt in KJ/mol are given below

	$IE_1 + IE_2$	$IE_3 + IE_4$
Ni	2.49	8.80
Pt	2.60	6.70

So, the correct statement is

- 1) Ni(II) compounds tend to be thermodynamically more stable than Pt(II)
 - 2) Pt(IV) compounds tend to be more stable than Ni(IV)
 - 3) 1 & 2 both
 - 4) None of these
11. VO_4^{3-} , CrO_4^{2-} and MnO_4^{2-} are pale yellow strong yellow and intense purple respectively in aqueous solution. The darkening colour is due to
- 1) Charge transfer
 - 2) d-d transition
 - 3) Half filled d-orbital shells
 - 4) Increasing number of unpaired electrons
12. The ratio of magnetic moments of Fe(III) & Co(II) is
- 1) $\sqrt{5} : \sqrt{7}$
 - 2) $\sqrt{35} : \sqrt{15}$
 - 3) 7 : 3
 - 4) $\sqrt{24} : \sqrt{15}$
13. Which is not an interstitial compound
- 1) Tin
 - 2) Fe_2O_3
 - 3) Mn_2C_3
 - 4) W_2C
14. The metal that has the highest melting point and used in making hard steel is
- 1) Cu
 - 2) Mn
 - 3) Zn
 - 4) W
15. The highest degree of paramagnetism is shown by
- 1) $CoCl_2 \cdot H_2O$
 - 2) $MnSO_4 \cdot 4H_2O$
 - 3) $FeCl_2 \cdot 4H_2O$
 - 4) $NiCl_2 \cdot 6H_2O$
16. The correct statement among the following is
- 1) The colour of $Cr_2O_7^{2-}$ ion is due to d-d transition of unpaired electrons
 - 2) Transition elements form a large number of alloys, because of similar boiling points
 - 3) Bronze is an alloy of Copper and Zinc
 - 4) Salt of Fe^{2+} ion has greenish colour
17. Which of the following is ferrous alloy?
- 1) German Silver
 - 2) Gun metal
 - 3) Nichrome
 - 4) Devarda's alloy
18. Which of the following is not the property of interstitial compound
- 1) They have high M.P, higher than those of pure metals
 - 2) They are less harder than diamond
 - 3) They retain metallic conductivity
 - 4) Their chemical reactivity is very high

Preparation and properties of compounds

19. From sodium Argentocyanide $\text{Na}[\text{Ag}(\text{CN})_2]$, silver is precipitated by adding of
1) Tin 2) Zinc 3) Mercury 4) Calcium
20. Which of the following ions is coloured ?
1) Cu^+ 2) Cu^{+2} 3) Ti^{+2} 4) V^{+3}
21. The reactivity of copper is low because of its
1) High enthalphy of sublimation and low ionization energy
2) High enthalphy of sublimation and high ionization energy
3) Low enthalphy of sublimation and high ionization energy
4) Low enthalphy of sublimation and low ionization energy
22. When K_2CrO_4 is added to CuSO_4 solution, there is formation of CuCrO_4 as well as CuCr_2O_7 . Formation of CuCr_2O_7 is due to
1) Basic nature of CuSO_4 solution 2) Acidic nature of CuSO_4 solution
3) CuSO_4 oxidizes CrO_4^{-2} to $\text{Cr}_2\text{O}_7^{-2}$ 4) This is the typical property of CuSO_4
23. Gold dissolves in aquaregia forming
1) $\text{Au}(\text{NO}_3)_2$ 2) AuCl_3 3) $\text{H}[\text{AuCl}_4]$ 4) AuNO_3
24. Which one is least soluble in water ?
1) Ag_2S 2) AgCl 3) AgI 4) AgBr
25. Which compound is deliquescent ?
1) Hg_2Cl_2 2) HgCl_2 3) ZnCl_2 4) CdCl_2
26. HgCl_2 is reduced to Hg_2Cl_2 by
1) CH_3COOH 2) CCl_4 3) HCOOH 4) NH_3
27. Lithopone is a mixture of
1) $\text{ZnSO}_4, \text{BaSO}_4$ 2) $\text{ZnS}, \text{BaSO}_4$ 3) $\text{ZnCO}_3, \text{BaCO}_3$ 4) $\text{ZnS}, \text{Na}_2\text{SO}_4$
28. $\text{AgF}, \text{AgCl}, \text{NaCl}, \text{NaBr}, \text{NaI}$ are colourless but AgBr and AgI are coloured because
1) Ag^+ polarizes Br^- and I^- and not able to polarize Cl^- and F^-
2) AgBr has unpaired electron
3) AgBr has defects 4) All of these
29. Which one of the following statements is correct ?
1) Hg_2Cl_2 is called corrosive sublimate
2) Hg_2Cl_2 gives white precipitate with ammonium hydroxide
3) Hg_2Cl_2 is used as a purgative
4) Hg_2Cl_2 is soluble in water

30. Which of the following statements is correct ?
- 1) FeI_3 does not exist in a pure state because Fe^{3+} oxidizes I^-
 - 2) Fe(III) complexes are less stable than Fe(II) complexes
 - 3) Fe^{+3} has a d^6 configuration
 - 4) Fe^{2+} reacts with the SCN^- ion to produce $[\text{Fe}(\text{SCN})(\text{H}_2\text{O})_5]^{2+}$
31. Given reagents are HCl , NaOH , ZnCl_2 , Na_2CO_3 , NH_4Cl and Zn . Which of these reagent(s) can intensify hydrolysis of FeCl_3 when added to its solution
- 1) NaOH , Na_2CO_3
 - 2) NaOH , NH_4Cl , HCl
 - 3) Na_2CO_3 , $\text{NaOH}+\text{HCl}$
 - 4) NH_4Cl , HCl , ZnCl_2
32. Zinc carbonate is precipitated from zinc sulphate solution by the addition of
- 1) Na_2CO_3
 - 2) CaCO_3
 - 3) MgCO_3
 - 4) NaHCO_3
33. Which one of the following statements is wrong ?
- 1) HgCl_2 dissolves in hot water
 - 2) HgCl_2 gives HCl when treated with sulphuric acid
 - 3) HgCl_2 gives yellow precipitate with NaOH
 - 4) HgCl_2 gives white precipitate with ammonium hydroxide
34. It is always advisable not to cover egg yolk or mustard with silver cutlery because
- 1) silver reacts with water of egg yolk to form AgOH
 - 2) silver reacts with sulphur of egg yolk forming black Ag_2S
 - 3) silver reacts with egg yolk forming Ag_2SO_4 which is a poisonous substance
 - 4) silver attracts UV light of the atmosphere, thereby spoiling the food
35. Silver halides are used in photography because they are
- 1) photosensitive
 - 2) soluble in hypo solution
 - 3) soluble in NH_4OH
 - 4) dilNH_4OH
36. When a reagent (A) reacts with FeCl_3 , the solution turns red due to the formation of a compound (B). The reagent causes no change in colour when it reacts with FeCl_2 in pure state. A and B respectively are
- 1) $\text{K}_4[\text{Fe}(\text{CN})_6]$ and $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
 - 2) NH_4SCN and $[\text{Fe}(\text{SCN})]^{2+}$
 - 3) $\text{K}_3[\text{Fe}(\text{CN})_6]$ and $\text{K}_2[\text{Fe}(\text{CN})_6]$
 - 4) NH_4CNS and $[\text{Fe}(\text{CNS})]^{2+}$
37. The strength as oxidizing agents of the following species in acidic medium decreases in the order
- 1) $\text{S}_2\text{O}_8^{2-} > \text{Cr}_2\text{O}_7^{2-} > \text{MnO}_4^-$
 - 2) $\text{Cr}_2\text{O}_7^{2-} > \text{S}_2\text{O}_8^{2-} > \text{MnO}_4^-$
 - 3) $\text{MnO}_4^- > \text{Cr}_2\text{O}_7^{2-} > \text{S}_2\text{O}_8^{2-}$
 - 4) $\text{S}_2\text{O}_8^{2-} > \text{MnO}_4^- > \text{Cr}_2\text{O}_7^{2-}$
38. MnO_4^- has the strongest and weakest oxidizing power in
- 1) alkaline and acidic medium
 - 2) alkaline and neutral medium
 - 3) acidic and basic medium
 - 4) acidic and neutral medium

Additional Questions :

39. Which of the following statements is incorrect ?

- 1) The compounds formed by the corresponding 3d transition metals in lower valence states are ionic but those in higher valence states are covalent.
- 2) The 4d and 5d transition metals form less ionic compounds than the 3d transition metals.
- 3) The compounds formed by the 3d transition metals are less ionic than the corresponding compounds formed by the 4d and 5d transition metals.
- 4) The ionization energies of 3d, 4d and 5d transition metals are greater than those of group 1 and 2 metals.

40. Which of the following statements is correct ?

- 1) The purple colour of the permanganate ion does not arise from d-d transition but from charge transfer.
- 2) The purple colour of the permanganate ion arises due to d-d transition.
- 3) The purple colour of permanganate ion arises due to both d-d transition and charge transfer.
- 4) For charge transfer to occur in the MnO_4^- ion the difference in energy between Mn^{7+} and O^{2-} should be very high.

41. Which of the following statement is correct ?

- 1) The second row elements have smaller radii than the corresponding third row ones.
- 2) Because of the lanthanide contraction, the radii of the third row elements are almost the same as those of the first elements.
- 3) Because of the lanthanide contraction, the radii of the third row elements are almost the same as those of the second row elements.
- 4) Because of the lanthanide contraction the separation of second row elements from one another is easier.

42. E° values for the couples $\text{Cr}^{3+} / \text{Cr}^{2+}$ and $\text{Mn}^{3+} / \text{Mn}^{2+}$ are -0.41 and $+1.51$ volts respectively. These values suggest that

- 1) Cr^{2+} acts as a reducing agent whereas Mn^{3+} acts as an oxidizing agent
- 2) Cr^{2+} is more stable than Cr^{3+} state
- 3) Mn^{3+} is more stable than Mn^{2+}
- 4) Cr^{2+} acts as an oxidizing agent whereas Mn^{3+} acts as a reducing agent.

43. Among the following series of transition metal ions the one where the metal ions have $3d^2$ electronic configuration is

- | | |
|--|--|
| 1) $\text{Ti}^{3+}, \text{V}^{2+}, \text{Cr}^{3+}, \text{Mn}^{4+}$ | 2) $\text{Ti}^{3+}, \text{V}^{4+}, \text{Cr}^{3+}, \text{Mn}^{7+}$ |
| 3) $\text{Ti}^{4+}, \text{V}^{3+}, \text{Cr}^{2+}, \text{Mn}^{3+}$ | 4) $\text{Ti}^{2+}, \text{V}^{3+}, \text{Cr}^{4+}, \text{Mn}^{5+}$ |

44. Four successive members of the first row transition elements are listed below with their atomic numbers. Which one of them is expected to have the highest third ionization enthalpy ?

- | | |
|----------------------|-----------------------|
| 1) Vanadium (Z = 23) | 2) Manganese (Z = 25) |
| 3) Chromium (Z = 24) | 4) Iron (Z = 26) |

45. When K_2CrO_4 is added to $CuSO_4$ solution, there is formation of $CuCrO_4$ as well as $CuCr_2O_7$. Formation of $CuCr_2O_7$ is due to
- 1) basic nature of $CuSO_4$ solution
 - 2) acidic nature of $CuSO_4$ solution
 - 3) $CuSO_4$ oxidizes CrO_4^{2-} to $Cr_2O_7^{2-}$
 - 4) this is the typical property of $CuSO_4$
46. The m.pt of transition elements is very high due to
- 1) Involvement of greater no. of electrons in (n-1)d and ns electrons in the inter atomic metallic bonding.
 - 2) Non-involvement of greater no. of electrons in (n-1)d and ns electrons in the inter-atomic metallic bonding.
 - 3) Involvement of greater no. of elements in 'ns' electrons in the inter-atomic metallic bonding
 - 4) Non-involvement of greater no. of electrons in 'ns' electrons in the inter-atomic metallic bonding.
47. Which of the 3d series transition elements exhibits the largest number of oxidation states
- 1) Sc
 - 2) V
 - 3) Mn
 - 4) Fe
48. The observed magnetic moment of Co^{2+} is much higher than the calculated magnetic moment by $\sqrt{n(n+2)}$ formula due to
- 1) The spin angular momentum
 - 2) The presence of more number of unpaired electrons than expected
 - 3) The consideration of orbital angular momentum along with spin angular momentum of electrons.
 - 4) The presence of more no. of paired electrons.
49. The catalyst used in the reduction of perdisulphate to sulphate with iodide ion is
- 1) Fe^{3+}
 - 2) Fe^{2+}
 - 3) Both (1) and (2)
 - 4) Cr^{2+}
50. A transition element M forms the oxides MO , M_2O_3 , MO_3 , and M_2O_7 . Which of the following statements about these oxides is true ?
- 1) MO is most acidic
 - 2) M_2O_3 is the most likely to be a strong oxidizing agent.
 - 3) MO_3 is the most acidic
 - 4) M_2O_7 is the one that cannot be a reducing agent
51. Amongst the following species, maximum covalent character is exhibited by
- 1) $FeCl_2$
 - 2) $ZnCl_2$
 - 3) $HgCl_2$
 - 4) $CdCl_2$
52. Arrange following ions in the decreasing order of their magnetic moment.
- I) V^{4+}
 - II) Mn^{4+}
 - III) Fe^{3+}
 - IV) Ni^{2+}
- [At nos. V = 23 Mn = 25 Fe = 26 Ni = 28]
- 1) II > III > I > IV
 - 2) III > IV > II > I
 - 3) III > II > IV > I
 - 4) I < IV < III < II
53. Which of the following pairs of transition metal ions are the stronger oxidizing agents in aqueous solutions.
- 1) V^{2+} and Cr^{2+}
 - 2) Ti^{2+} and Cr^{2+}
 - 3) Mn^{3+} and Co^{3+}
 - 4) V^{2+} and Fe^{2+}
54. Lanthanides like Eu and Yb can form hydrides like EuH_2 and YbH_2 . These hydrides are
- 1) Ionic hydrides
 - 2) Covalent hydrides
 - 3) Metallic hydrides
 - 4) Intermediate hydrides

55. The electronic configuration of actinoides cannot be assigned with degree of certainty because of
 1) Overlapping of inner orbitals
 2) Free moment of electrons over all the orbital's movement
 3) Small energy difference between 5f and 6d levels
 4) None of the above
56. Number of moles of $K_2Cr_2O_7$ reduced by one mole of Sn^{+2} ion is
 1) $\frac{1}{3}$ 2) 3 3) $\frac{1}{6}$ 4) 6
57. Which of the following ions is coloured ?
 1) $_{57}La^{3+}$ 2) $_{63}Eu^{3+}$ 3) $_{64}Gd^{3+}$ 4) $_{71}Lu^{3+}$
58. Cerium can show the oxidation state of +4 because
 1) It resembles alkali metals
 2) It has very low value of ionization energy
 3) It has a tendency to attain noble gas configuration of xenon
 4) It has a tendency to attain f^0 configuration
59. Among the following the compound that is both paramagnetic and coloured ?
 1) $K_2Cr_2O_7$ 2) $(NH_4)_2[TiCl_6]$ 3) $CoSO_4$ 4) $K_3[Cu(CN)_4]$
60. Arrange in increasing acidic nature
 I) $[Na(H_2O)_6]^+$ II) $[Sc(H_2O)_6]^{3+}$ III) $[Mn(H_2O)_6]^{2+}$ IV) $[Ni(H_2O)_6]^{2+}$
 1) $IV > III > II > I$ 2) $I > II > III > IV$ 3) $II > IV > III > I$ 4) $IV > II > III > I$

Numerical Value Type Questions

61. A metal ion from the first transition has a magnetic moment of 3.87 B.M. How many unpaired electrons are expected to be present in the ion
62. The brown ring complex compound is formulated as $[Fe(H_2O)_5(NO)]SO_4$. The oxidation state of Iron is +x
63. The covalency of Mg^{2+} ion in chlorophyll is
64. How many moles of $KMnO_4$ are required to oxidise 10 moles of ferrous oxalate in acidic medium.
65. When potassium dichromate is heated to a white heat, it decomposes to give potassium chromate, chromic oxide and oxygen gas. The number of moles of oxygen gas liberated in the decomposition process are

KEY SHEET

EXERCISE-I

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1) 3 | 2) 2 | 3) 4 | 4) 2 | 5) 2 | 6) 3 | 7) 2 | 8) 1 | 9) 1 | 10) 2 |
| 11) 2 | 12) 4 | 13) 4 | 14) 1 | 15) 2 | 16) 4 | 17) 3 | 18) 4 | 19) 2 | 20) 1 |
| 21) 3 | 22) 2 | 23) 2 | 24) 4 | 25) 2 | 26) 1 | 27) 1 | 28) 3 | 29) 2 | 30) 2 |
| 31) 4 | 32) 2 | 33) 2 | 34) 1 | 35) 2 | 36) 1 | 37) 4 | 38) 1 | 39) 3 | 40) 1 |
| 41) 2 | 42) 1 | 43) 1 | 44) 1 | 45) 2 | 46) 3 | 47) 1 | 48) 1 | 49) 1 | 50) 3 |

51) 4 52) 3 53) 3 54) 1 55) 3 56) 4 57) 3 58) 4 59) 8 60) 6
 61) 6 62) 3 63) 0 64) 8

EXERCISE-II

1) 3 2) 1 3) 3 4) 1 5) 3 6) 3 7) 1 8) 4 9) 3 10) 3
 11) 1 12) 2 13) 2 14) 4 15) 2 16) 4 17) 3 18) 4 19) 2 20) 2
 21) 2 22) 2 23) 3 24) 1 25) 3 26) 3 27) 2 28) 1 29) 3 30) 1
 31) 1 32) 4 33) 2 34) 2 35) 1 36) 2 37) 4 38) 3 39) 3 40) 1
 41) 3 42) 1 43) 4 44) 2 45) 2 46) 1 47) 3 48) 3 49) 1 50) 4
 51) 3 52) 3 53) 3 54) 1 55) 3 56) 1 57) 2 58) 4 59) 3 60) 3
 61) 3 62) 1 63) 4 64) 6 65) 1

LEVEL-II (ADVANCED)**More than One correct answer Type Questions**

- Which of the following is/are characteristic(s) of the transition elements in the series from scandium to copper ?
 - The formation of coloured cations
 - The presence of at least one unpaired electron in a d-orbital of a cation
 - The ability to form complex ions
 - The possession of an oxidation state of +1
- Which of the following represent(s) the correct sequence of indicated property ?
 - $\text{Sc}^{3+} > \text{Fe}^{3+} > \text{Mn}^{3+}$; Stability of +3 oxidation state
 - $\text{Mn}^{2+} < \text{Ni}^{2+} < \text{Co}^{2+} < \text{Fe}^{2+}$; magnetic moment
 - $\text{FeO} > \text{CoO} > \text{NiO}$; basic character
 - $\text{Sc} < \text{Ti} < \text{V} < \text{Cr}$ number of oxidation states
- Among the following the correct statement is/are
 - In 6th (VIA) transition group, the lower oxidation state is favoured by heavier elements (Mo and W) when compared to chromium
 - Cu^{2+} is more stable than $\text{Cu}^{+}_{(\text{aq})}$
 - E^0 of Zn^{2+}/Zn is more negative than expected because of extra stability of d^{10} configuration
 - In 3d-series Mn^{3+} and Co^{3+} are the strongest oxidizing agents when only +3 oxidation state is considered
- The correct statements about the transition elements is/are
 - The most stable oxidation state is +2 and its stability decreases across the period
 - Transition elements of 3d series have almost same atomic sizes from Cr to Cu
 - the stability of +2 oxidation state increases across the period
 - Some transition elements like Ni, Fe, Cr may show zero oxidation state in some of their compounds

5. Which are the true statements regarding ionization energy of d series elements ?
- Ionization energy of Mn is greater than Cr
 - Ionization energy of Pd is greater than Rh and Ag
 - Ionization energy decreases from 3d to 4d transition series but it increases from 4d to 5d series
 - Ionization energy of Mo is greater than that of W
6. Which of the following compounds is expected to be coloured :
- Ag_2SO_4
 - CuF_2
 - MnCl_2
 - CuCl
7. Which of the following statement(s) is (are) correct. When a mixture of NaCl and $\text{K}_2\text{Cr}_2\text{O}_7$ is gently warmed with conc. H_2SO_4 :
- A deep red vapours is evolved
 - The vapours when passed into NaOH solution gives a yellow solution of Na_2CrO_4
 - Chlorine gas is evolved
 - Chromyl chloride is formed
8. K_2MnO_4 is formed when :
- Cl_2 is passed into KMnO_4 solution
 - Manganese dioxide is fused with potassium hydroxide in air
 - Formaldehyde reacts with potassium permanganate in presence of a strong alkali
 - Potassium permanganate reacts with conc. H_2SO_4
9. KMnO_4 is iso structural with
- KClO_4
 - HClO_4
 - K_3BO_3
 - None
10. Pick out the correct statements
- MnO_2 dissolves in conc. HCl but does not form Mn^{+4} ions
 - Decomposition of acidic KMnO_4 is not catalysed by sunlight
 - MnO_4^{2-} is strongly oxidising and stable only in very strong alkali. In dilute alkali water or acidic solutions it disproportionates
 - KMnO_4 does not act as oxidizing agent in alkali medium
11. Which of the following are correct about actinoids ?
- They show large number of oxidation states
 - They show actinoid contraction
 - They form oxo-cations
 - All of the are radioactive
12. Which of the following characteristics of lanthanides is correct ?
- Reducing property of lanthanides decreases from La to Lu
 - Complex forming ability of lanthanides increases from La to Lu
 - Basic character of the oxides and hydroxides increases from La to Lu
 - Lanthanides form stable compounds only in + 3 oxidation state

13. Which of the following statements concerning lanthanide elements is correct ?
- Lanthanides are separated from one another by an exchange method
 - Ionic radii of trivalent lanthanides steadily increases with increase in the atomic number
 - All lanthanides are dense metals
 - More characteristic oxidation state of lanthanide elements is + 3
14. Identify the incorrect statement among the following ?
- Shielding power of 4f electrons is quite weak
 - There is a decrease in the radii of the atoms or ions while proceeding from La to Lu
 - Lanthanoid contraction is the accumulation of successive shrinkages
 - As a result of lanthanoid contraction, the properties of 4d series of the transition elements have no similarities with 5d series of elements
15. The first ionization enthalpy of 5d series elements are highest as compared to 3d and 4d elements because
- Weak shielding of nucleus by 5f-electrons
 - Weak shielding of nucleus by 4f-electrons
 - Strong shielding of nucleus by 5d electrons
 - Strong shielding of nucleus by 4d-electrons
16. Consider the following statements
- The basic strength of hydroxides of lanthanides decreases from $\text{La}(\text{OH})_3$ to $\text{Lu}(\text{OH})_3$
 - The lanthanoid ions Lu^{3+} , Yb^{2+} and Ce^{4+} are diamagnetic
 - The correct order of ionic radii of Ce^{3+} , La^{3+} , Pm^{3+} and Yb^{3+} is $\text{Yb}^{3+} < \text{Pm}^{3+} < \text{La}^{3+} < \text{Ce}^{3+}$
- Which of the statements given above is/are correct ?
- 1 only
 - 1,2 only
 - 1, 2 and 3
 - None of these

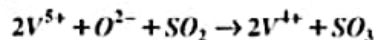
Linked Comprehension Type Questions

Passage - I :

Transition metal and their compounds are used as catalysts in industry and in biological system. For example, in the Contact Process, vanadium compounds in the + 5 state (V_2O_5 or VO_3^-) are

used to oxidise SO_2 to SO_3 : $\text{SO}_2 + \frac{1}{2}\text{O}_2 \xrightarrow{\text{V}_2\text{O}_5} \text{SO}_3$

It is thought that the actual oxidation process takes place in two stages. In the first step, V^{5+} in the presence of oxide ions converts SO_2 to SO_3 . At the same time, V^{5+} is reduced to V^{4+} .



In the second step, V^{5+} is regenerated from V^{4+} by oxygen : $2\text{V}^{4+} + \frac{1}{2}\text{O}_2 \rightarrow 2\text{V}^{5+} + \text{O}^{2-}$

The overall process is, of course, the sum of these two steps : $\text{SO}_2 + \frac{1}{2}\text{O}_2 \longrightarrow \text{SO}_3$

17. Transition metals and their compounds catalyse reactions because :
- they have completely filled s-subshell
 - they have a comparable size due to poor shielding of d-subshell
 - they introduce an entirely new reaction mechanism with a lower activation energy
 - they have variable oxidation states differ by two units

18. During the course of the reaction :
- catalyst undergoes changes in oxidation state
 - catalyst increases the rate constant
 - catalyst is regenerated in its original form when the reactants form the products
 - all are correct
19. Catalytic activity of transition metals depends on :
- their ability to exist in different oxidation states
 - the size of the metal atoms
 - the number of empty atomic orbitals available
 - none of these
20. Which of the following ion involved in the above process will show paramagnetism ?
- V^{5+}
 - V^{4+}
 - O^{2-}
 - VO_3^-

Passage - II :

MnO₂ is the most important oxide of manganese. MnO₂ occurs naturally as the black coloured mineral pyrolusite. It is an oxidising agent, and decomposes to Mn₃O₄ on heating to 530° C. It is used in the preparation of potassium permanaga-nate and in the production of Cl₂ gas. Over half a million tonnes per year of MnO₂ is used in dry batteries

21. In the laboratory, MnO₂ is made by :
- heating Mn in O₂
 - oxidising Mn²⁺ in air
 - electrolytic oxidation of MnO₄
 - Precipitating MnO₂ from solution when performing titration of KMnO₄ in alkaline medium
22. When MnO₂ is fused with KOH in the presence of air, the product formed is :
- purple colour KMnO₄
 - green colour K₂ MnO₄
 - colourless MnO₄
 - purple colour K₂MnO₄
23. MnO₂ dissolves in concentrated HCl to form :
- Mn⁴⁺ ion and Cl₂
 - Mn²⁺ ion and Cl₂
 - [MnCl₂]⁻² and Cl₂
 - only [MnCl₄]

Passage - III :

The f-block consists of the two series lanthanoids and actinoids. Because Lanthanum closely resembles the Lanthanoids. The Lanthanoid resemble one another more closely than do the members of ordinary transition elements in any series. They have only one stable oxidation state. The chemistry of the actinoids is on the other hand, much more complicated. The complication arises partly owing to the occurrence of a wide range of oxidation states in these elements and partly because their radioactivity creates special problems in their study

24. Wrong statement of the following is
- Atomic radius of Lanthanoids decreases steadily from Ce to Lu
 - The shielding of a 4f electron by another is less than one d-electron by another with increase in nuclear charge along the series
 - Identical radii of Zr and Hf is a consequence of Lanthanoid contraction
 - Ce^{+4} is a strong oxidising agent
25. Which of the following is non a use of Lanthanoids
- Misch metal is an alloy of Lanthanoid metal used to produce bullets, shells
 - Oxides of Lanthanoids can be used as catalysts in petroleum cracking
 - Oxides of some Lanthanoids are used as phosphors in television screens
 - All Lanthanoids are good oxidising agents in +3 state
26. i) Common oxidation state of actinoids is + 3 (Except thorium)
 ii) Maximum O.S of actinoids is + 7
 iii) Actinoids evolve H_2 with alkali's
 iv) configuration of thorium is $[\text{Rn}] 5f^0 6d^2 7s^2$
 Correct statements are
- i, iii
 - i, ii, iii
 - i, ii, iv
 - All are correct

Matrix Matching Type Questions

27. **Column - I**
(Alloys)

- Fe, Cr, V
- Cu, Zn & Ni
- Cu, Sn & Zn
- Fe, Cr & Ni

Column-II
(Composition)

- Chrome steel
- German silver
- Gun metal
- Stainless steel

28. **Column - I**

- K_2MnO_4
- KMnO_4
- $\text{K}_2\text{Cr}_2\text{O}_7$
- K_2CrO_4

Column - II

- Transition element in +6 state
- Oxidising agent in acid medium
- manufactured from pyrolusite ore
- manufactured from chromite ore

29. **Column - I (Element)**

- Pr
- Gd^{+2}
- Eu
- Tm^{+3}

Column - II (Configuration)

- $4f^{12}$
- $4f^3 6s^2$
- $4f^7 6s^2$
- $4f^7 5d^1$

Integer Type Questions

30. How many electrons are involved in reduction of KMnO_4 in basic reduction?
31. Number of Cr–O sigma bonds in dichromate in $\text{Cr}_2\text{O}_7^{2-}$ is
32. Highest oxidation state exhibited by as actinoid is
33. Rare earth metals belongs to the column of modern periodic table
34. Number of electrons in outer most shell of Ce ($Z = 58$)

KEY SHEET

- 1) abc 2) acd 3) bcd 4) abcd 5) abc 6) bc 7) abcd 8) bc 9) ab 10) ac
11) abcd 12) abd 13) acd 14) d 15) b 16) b 17) c 18) d 19) a 20) b
21) d 22) b 23) b 24) a 25) d 26) c 27) A-p; B-q; C-r; D-s
28) A-pqr; B-qr; C-pqs; D-pqs 29) A-q; B-s; C-r; D-p 30) 1 31) 8 32) 7
33) 3 34) 2

